Data Communication
CSE 2211

Chapter 5 : Analog Transmission (Digital to Analog)

5-1 DIGITAL-TO-ANALOG CONVERSION

- *Converting digital data to a bandpass analog signal is traditionally called digital-to-analog conversion*
- *It is the process of changing one of the characteristics of an analog signal (carrier signal) based on the information in digital data.*

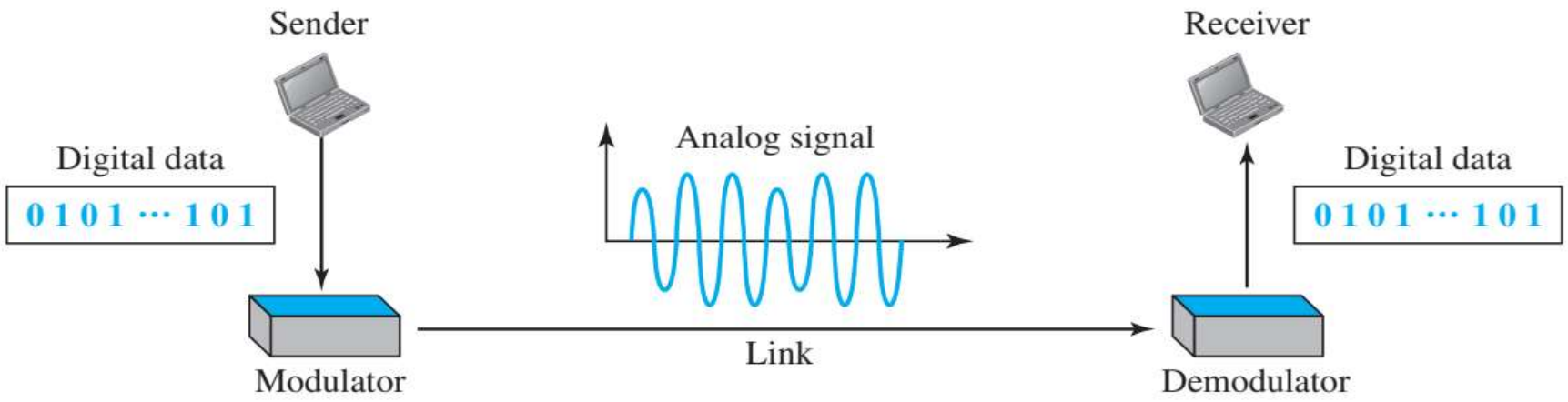
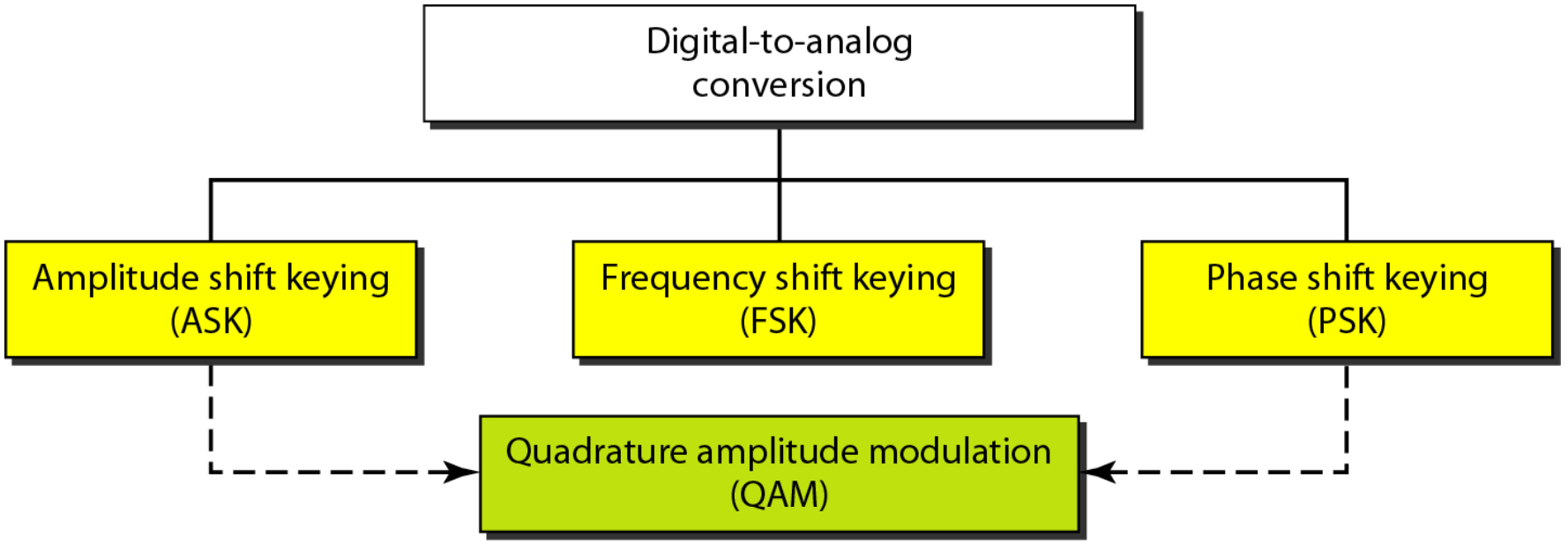


Figure 5.2 *Types of digital-to-analog conversion*



Note

Bit rate, N, is the number of bits per second (bps). Baud rate is the number of signal elements per second (bauds). In the analog transmission of digital data, the signal or baud rate is less than or equal to the bit rate.

$$S = N \times 1/r \text{ bauds}$$

Where r is the number of data bits per signal element.

Example 5.1

An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second, find the bit rate.

Solution

In this case, r = 4, S = 1000, and N is unknown. We can find the value of N from

$$S = N \times \frac{1}{r} \qquad \text{or} \qquad N = S \times r = 1000 \times 4 = 4000 \text{ bps}$$

Example 5.2

An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements do we need?

Solution

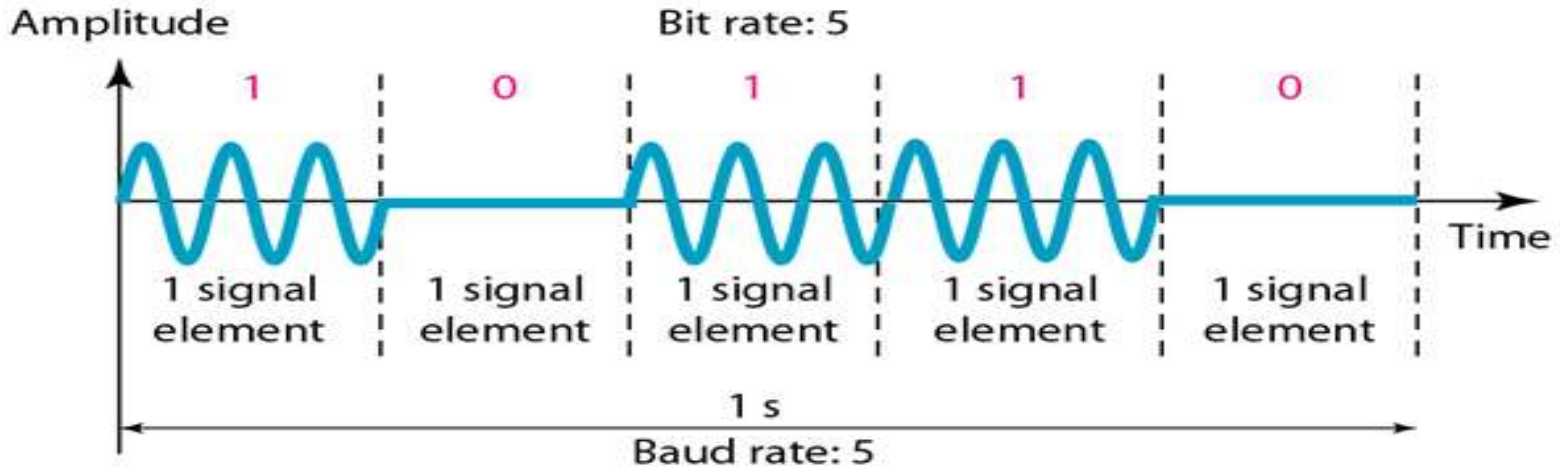
In this example, S = 1000, N = 8000, and r and L are unknown. We find first the value of r and then the value of L.

$$S = N \times \frac{1}{r} \rightarrow r = \frac{N}{S} = \frac{8000}{1000} = 8 \text{ bits/baud}$$

$$r = \log_2 L \rightarrow L = 2^r = 2^8 = 256$$

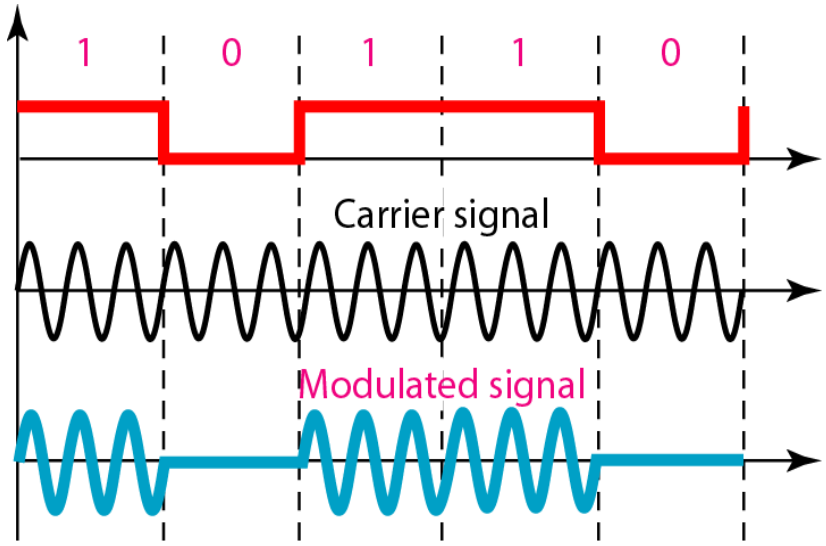
Amplitude Shift Keying (ASK)

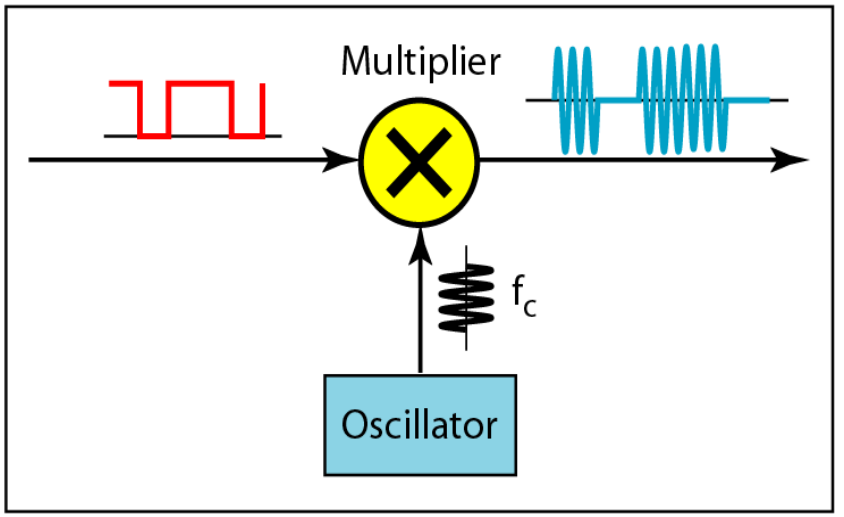
- ASK is implemented by changing the amplitude of a carrier signal to reflect amplitude levels in the digital signal.
- Both frequency and phase remains constant while the amplitude changes.
- In binary ASK, a digital "1" could not affect the signal, whereas a digital "0" would, by making it zero.
- Bit duration is the period of time that defines 1 bit.
- The peak amplitude during each bit duration is constant.



Implementation of BASK

- If digital data are presented as a unipolar NRZ digital signal with a high voltage of 1 V and a low voltage of 0 V, the implementation can achieved by multiplying the NRZ digital signal by the carrier signal coming from an oscillator.
- When the amplitude of the NRZ signal is 1, the amplitude of the carrier frequency is held;
- when the amplitude of the NRZ signal is 0, the amplitude of the carrier frequency is zero.





Bandwidth of ASK

- The bandwidth B of ASK is proportional to the signal rate S.

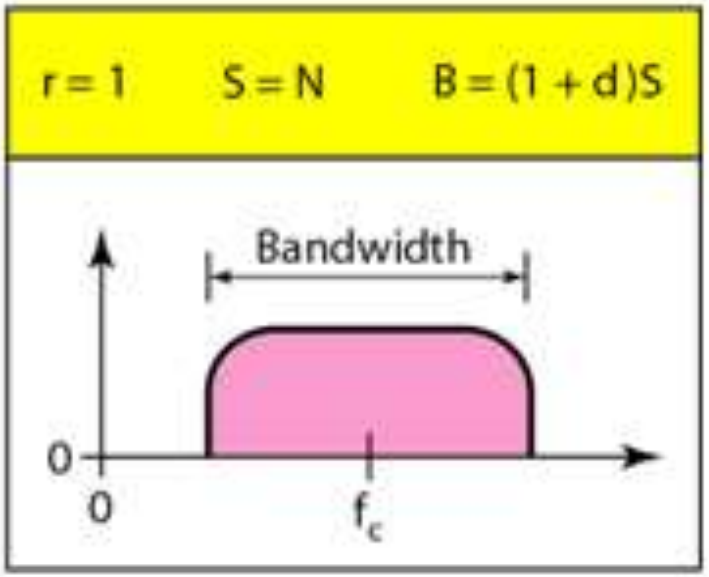
$$B = (1+d)S$$

- "d" is due to modulation and filtering, lies between 0 and 1.

Location of Bandwidth

If we have a bandpass channel available, we can choose our carrier frequency f_c in middle point so that the modulated signal occupies that bandwidth.

However, the most significant one are those between $f_c - S/2$ *And* $f_c + S/2$ *with* f_c *at the middle.*



Example 5.3

We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with d = 1?

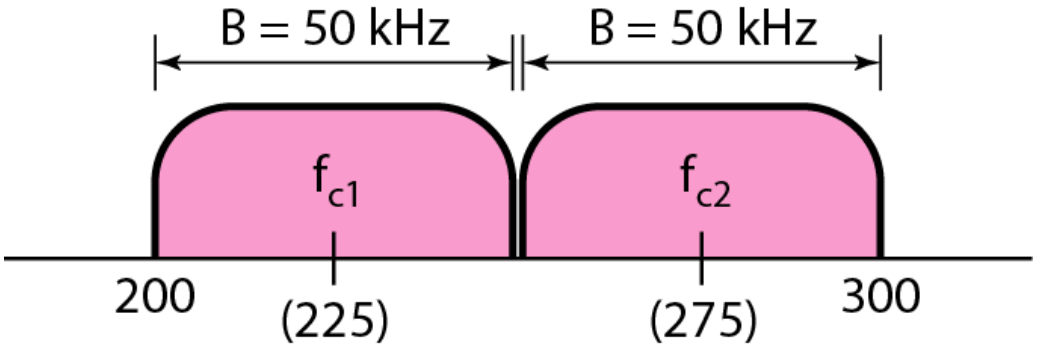
Solution

The middle of the bandwidth is located at 250 kHz. This means that our carrier frequency can be at $f_c = 250$ kHz. We can use the formula for bandwidth to find the bit rate (with d = 1 and r = 1).

$$B = (1 + d) \times S = 2 \times N \times \frac{1}{r} = 2 \times N = 100 \text{ kHz} \quad \rightarrow \quad N = 50 \text{ kbps}$$

Example 5.4

In data communications, we normally use full-duplex links with communication in both directions. We need to divide the bandwidth into two with two carrier frequencies, as shown in Figure. The figure shows the positions of two carrier frequencies and the bandwidths. The available bandwidth for each direction is now 50 kHz, which leaves us with a data rate of 25 kbps in each direction.



Frequency Shift Keying (FSK)

- In FSK, the frequency of the carrier signal is varied to represent binary 1 and 0.
- The frequency of the signal during each duration is constant, but changes for the next signal element if the data element changes.
- Both peak amplitude and phase remains constant.
- In Binary FSK (or BFSK), we can consider two carrier frequencies, f_1 and f_2.
- For example, a "1" could be represented by $f_1=f_c+\Delta f$, and a "0" could be represented by $f_2=f_c-\Delta f$.

Implementation of BFSK

- There are two implementations of BFSK: noncoherent and coherent.
- In noncoherent BFSK, there may be discontinuity in the phase when one signal element ends and the next begins.
- In coherent BFSK, the phase continues through the boundary of two signal elements.

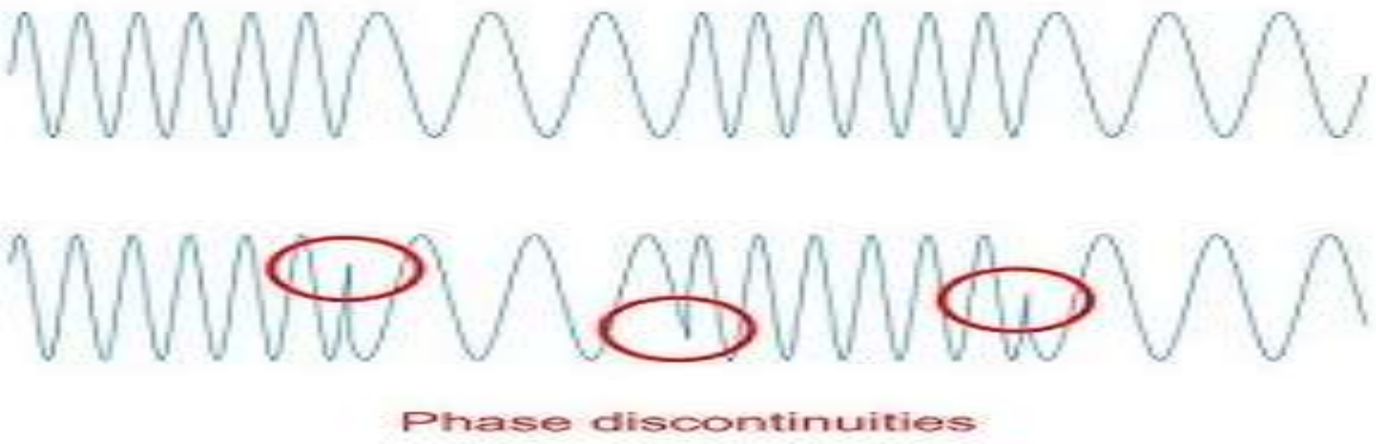
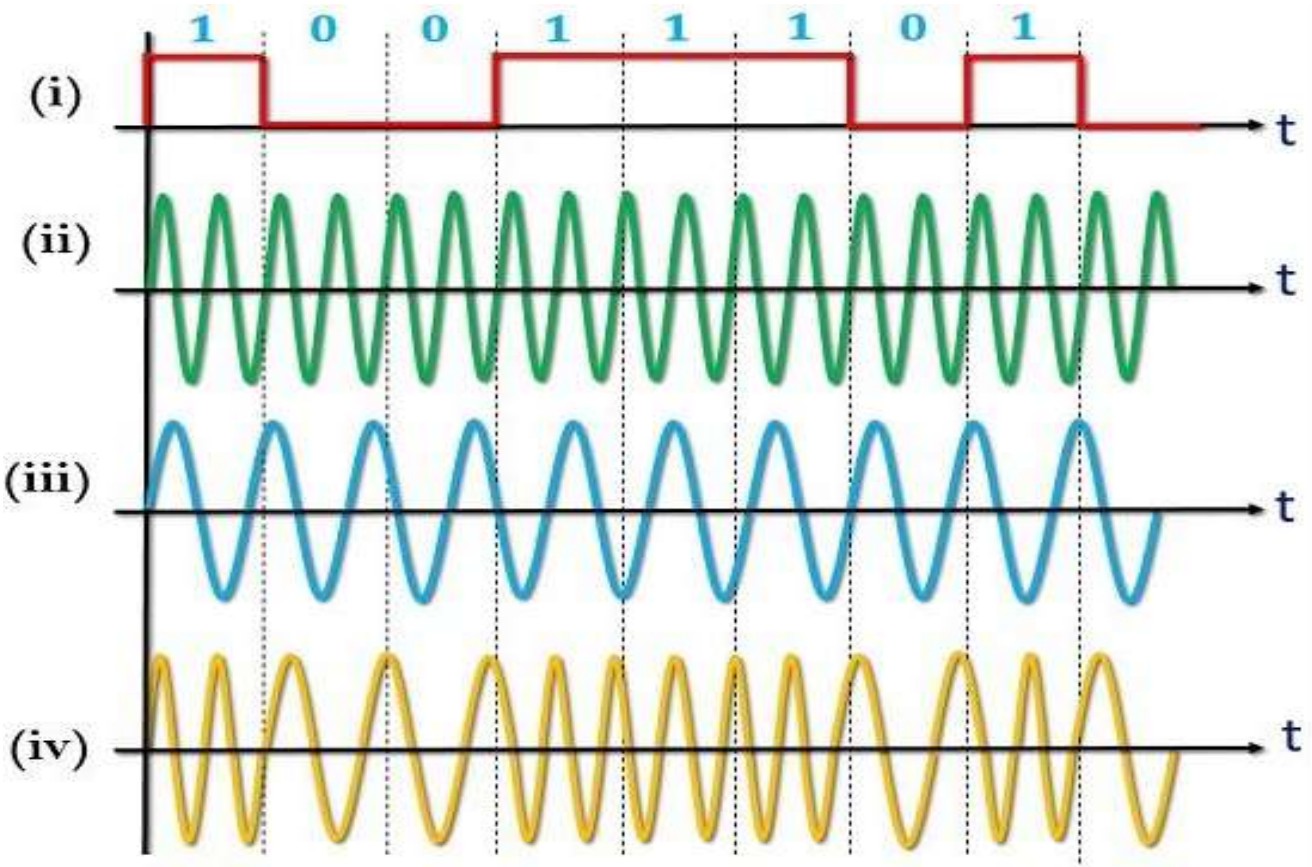


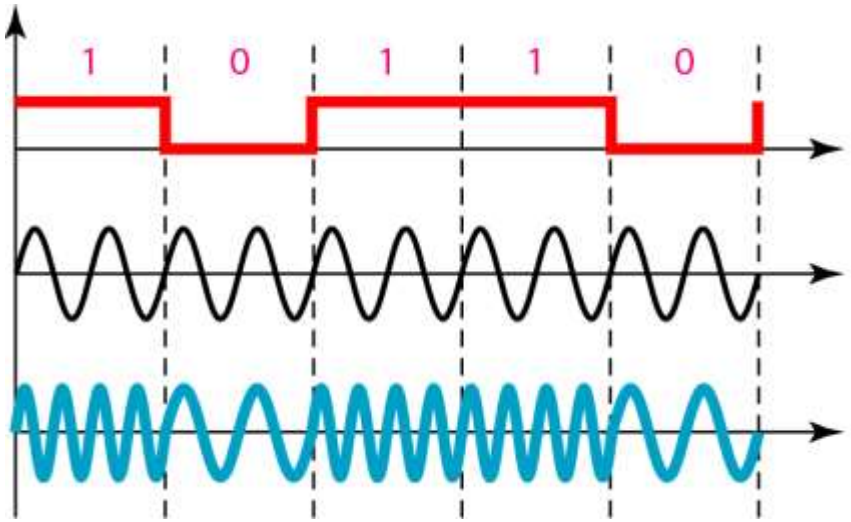
Figure Non coherent *Binary frequency shift keying*

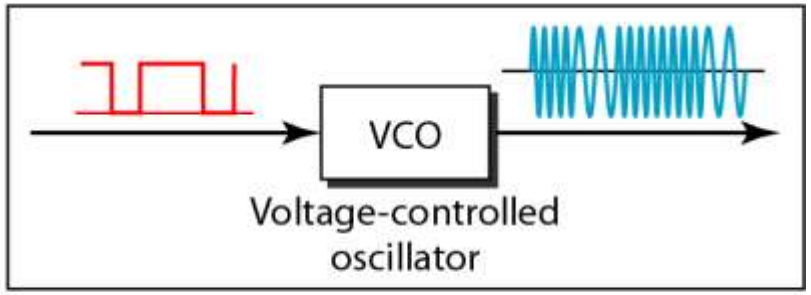


(i) Digital bitstream
(ii) High frequency carrier wave f_1
(iii) Low frequency carrier wave f_2
(iv) FSK modulated wave

Implementation of Coherent BFSK

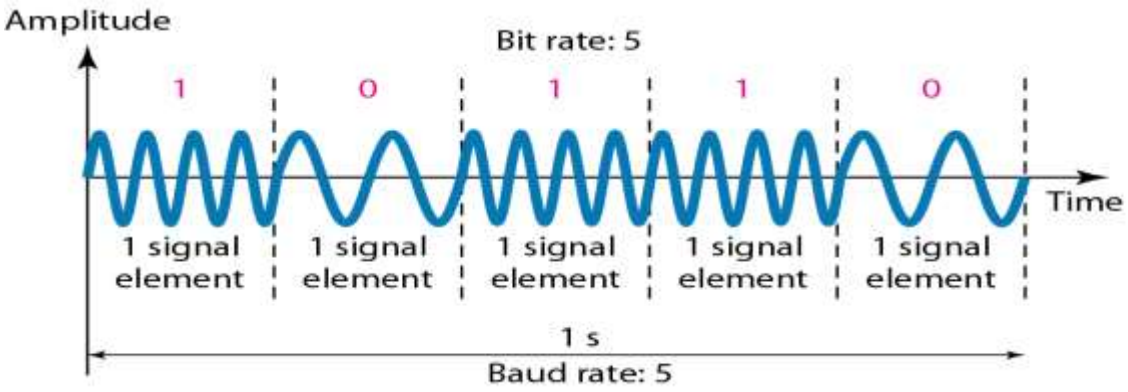
- The input to the oscillator is the unipolar NRZ signal.
- When the amplitude of NRZ is zero, the oscillator keeps its regular frequency;
- when the amplitude is positive, the frequency is increased.

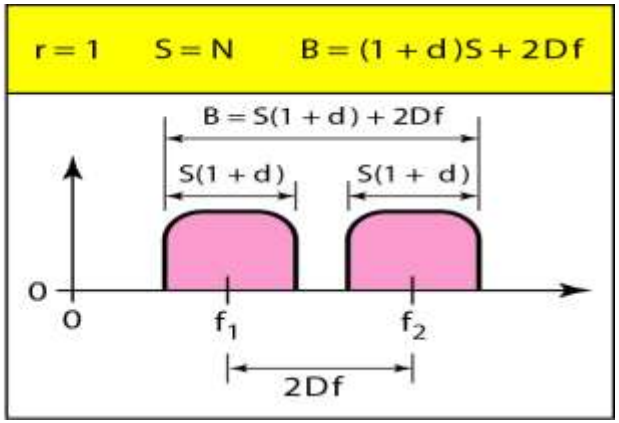




Bandwidth of FSK

- Carrier signals are only simple sine waves, but the modulation creates a nonperiodic composite.
- Two carrier frequencies: f_1 and f_2, with difference of $2\Delta f$
- Bandwidth at each carrier frequency is also $(1+d)\ S$
- Required bandwidth of BFSK is $B = (1+d)\ S + 2\Delta f$
- Min. frequency difference: $2\Delta f = (1+d)\ S$??
- It can be shown that min. of 2Δf should be at least S for the proper operation of modulation and demodulation.





Example 5.5

We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What should be the carrier frequency and the bit rate if we modulated our data by using FSK with d = 1?

Solution

This problem is similar to Example 5.3, but we are modulating by using FSK. The midpoint of the band is at 250 kHz. We choose 2Δf to be 50 kHz; this means

$$B = (1 + d) \times S + 2\Delta f = 100 \rightarrow 2S = 50 \text{ kHz} \quad S = 25 \text{ kbaud} \quad N = 25 \text{ kbps}$$

Multi level FSK

- We can use more than two frequencies.
- We can use four different frequencies f_1, f_2, f_3, and f_4 to send 2 bits at a time. To send 3 bits at a time, we can use eight frequencies.
- However, we need to remember that the frequencies need to be $2\Delta f$ apart.
- For the proper operation of the modulator and demodulator, it can be shown that the minimum value of $2\Delta f$ needs to be S.
- We can show that the bandwidth with d = 0 is

$$B = (1+d)xS + (L-1)2\Delta f = LxS$$

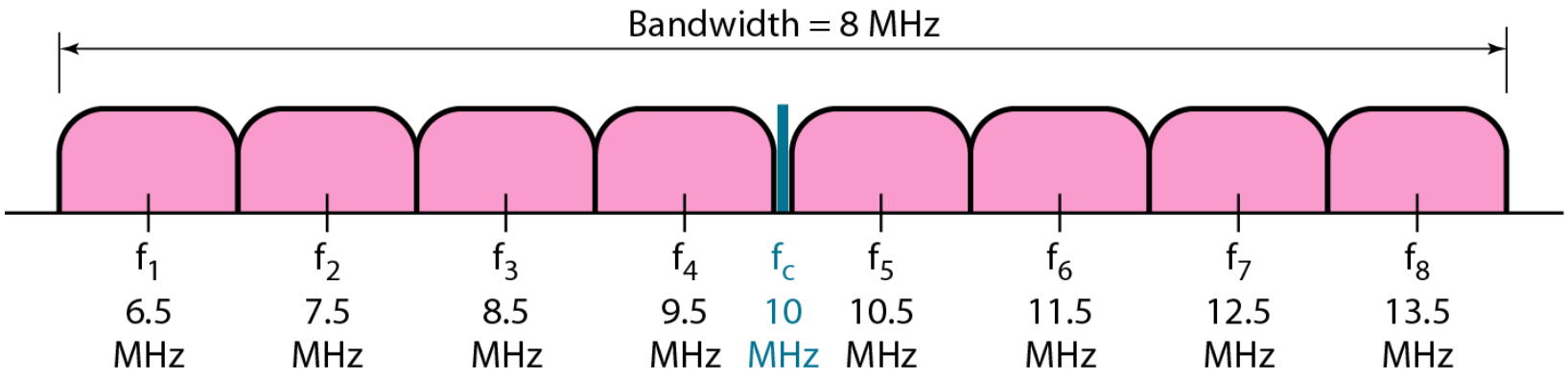
Example 5.6

We need to send data 3 bits at a time at a bit rate of 3 Mbps. The carrier frequency is 10 MHz. Calculate the number of levels (different frequencies), the baud rate, and the bandwidth.

Solution

We can have $L = 2^3 = 8$. The baud rate is $S = 3$ Mbps/3 = 1 Mbaud. This means that the carrier frequencies must be 1 MHz apart ($2\Delta f = 1$ MHz). The bandwidth is $B = 8 \times 1M = 8$MHz. Figure 5.8 shows the allocation of frequencies and bandwidth.

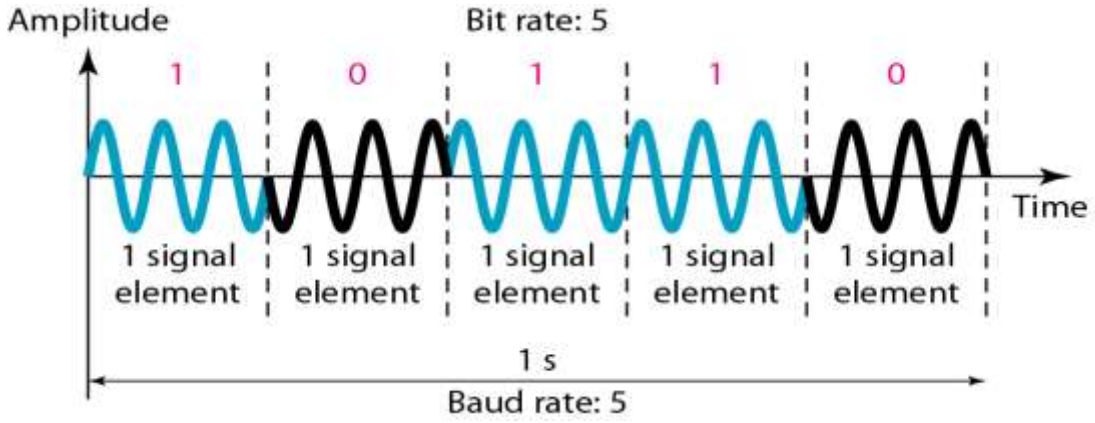
Figure 5.8 *Bandwidth of MFSK used in Example 5.6*

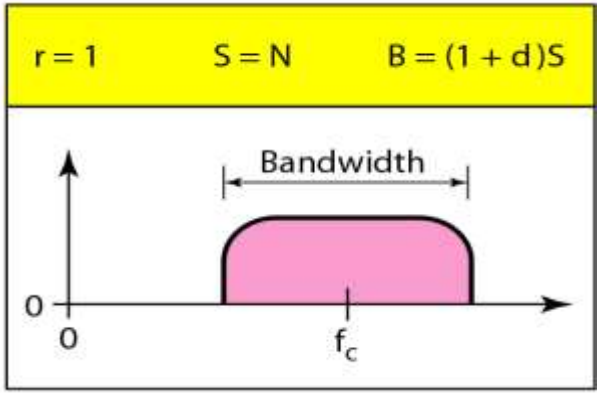


Phase Shift Keying

- The phase of the carrier is varied to represent two or more different signal elements.
- In binary PSK, we have only two signal elements, one with a phase of 0°, and the other with a phase of 180°.
- Both peak amplitude and frequency remain constant
- The bandwidth requirement, B is:

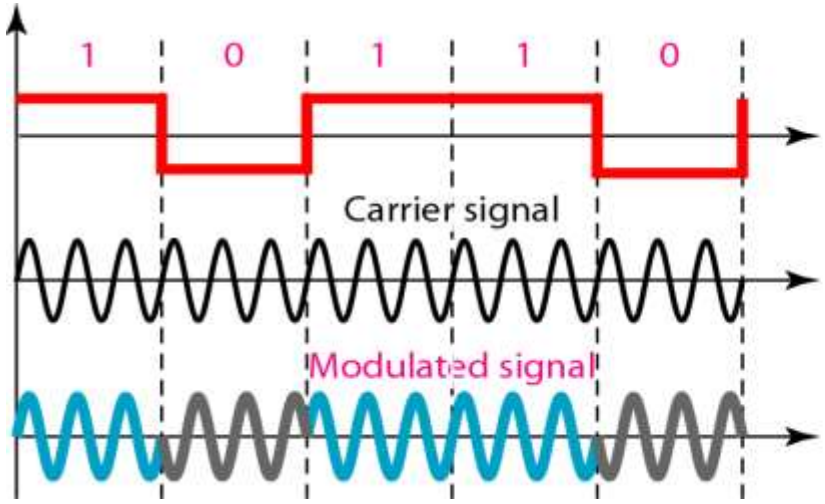
$$B = (1+d)xS$$

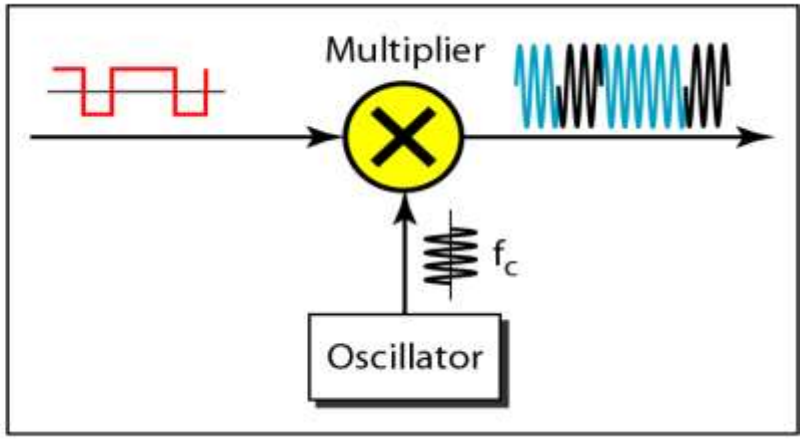




Implementation of BPSK

- We use the same idea we used for ASK but with a polar NRZ signal instead of a unipolar NRZ signal;
- The polar NRZ signal is multiplied by the carrier frequency;
- The 1 bit (positive voltage) is represented by a phase starting at 0°; the 0 bit (negative voltage) is represented by a phase starting at 180°.

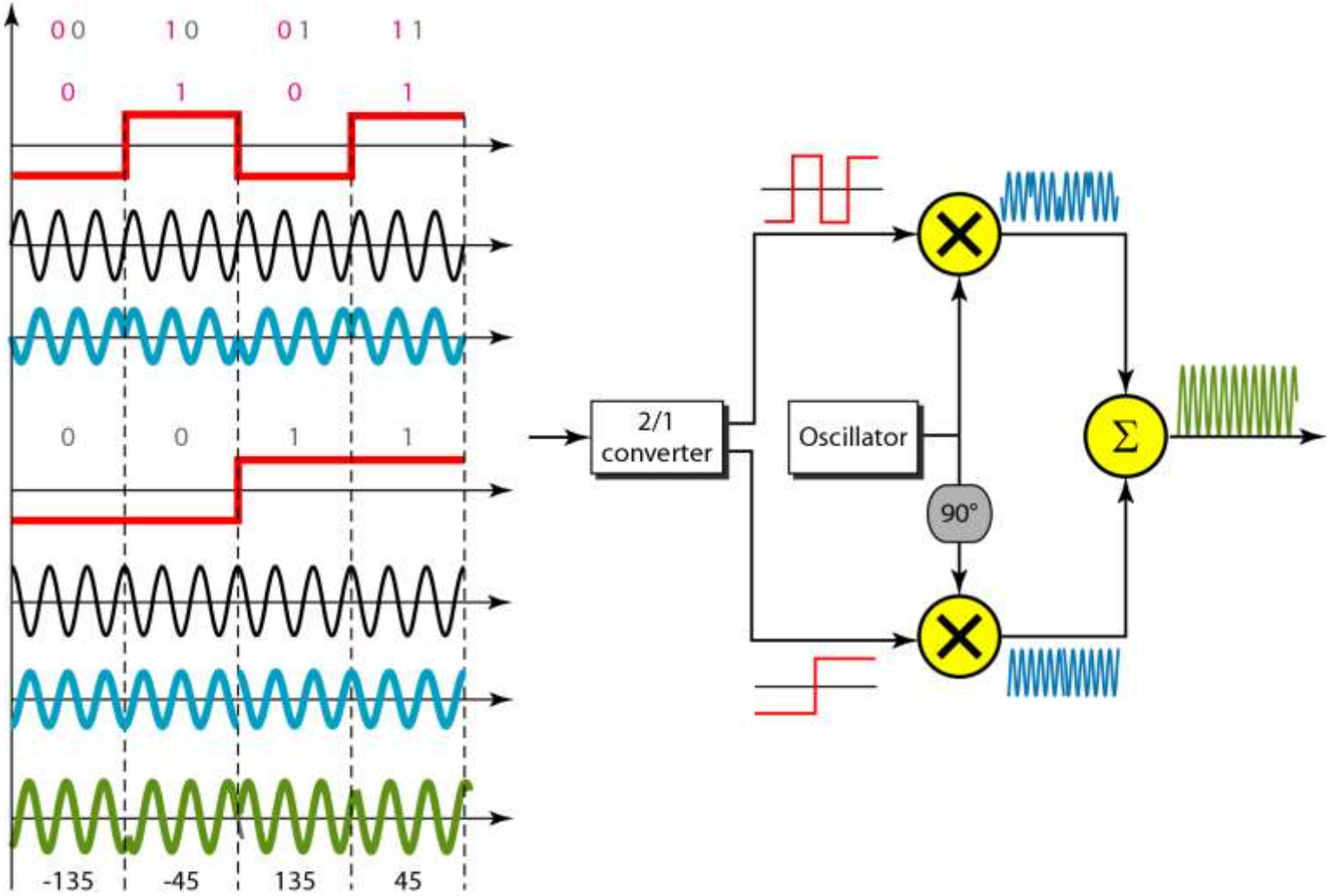




Quadrature PSK

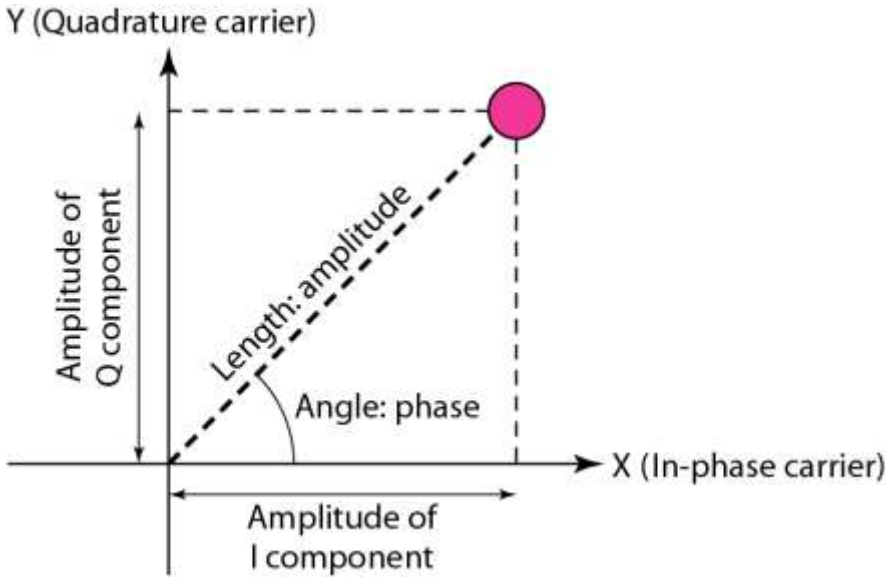
- To increase the bit rate, we can code 2 or more bits onto one signal element.
- In QPSK, we parallelize the bit stream so that every two incoming bits are split up and PSK a carrier frequency. One carrier frequency is phase shifted 90° from the other - in quadrature.
- The two PSKed signals are then added to produce one of 4 signal elements. L = 4 here.

Figure 5.11 *QPSK and its implementation*



Constellation Diagram

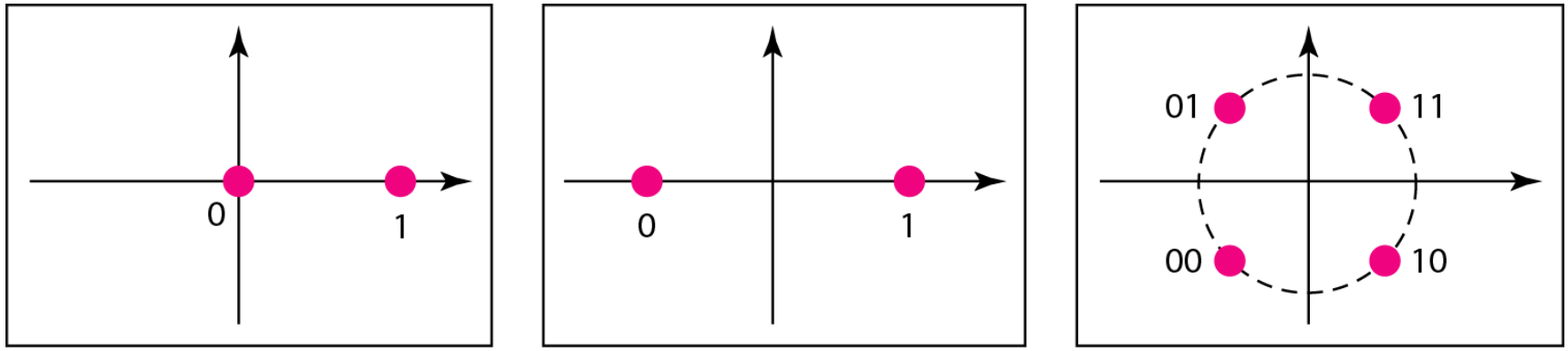
- *A constellation diagram helps us to define the amplitude and phase of a signal when we are using two carriers, one in quadrature of the other.*
- *The X-axis represents the in-phase carrier and the Y-axis represents quadrature carrier.*



Example

Show the constellation diagrams for an ASK (OOK), BPSK, and QPSK signals.

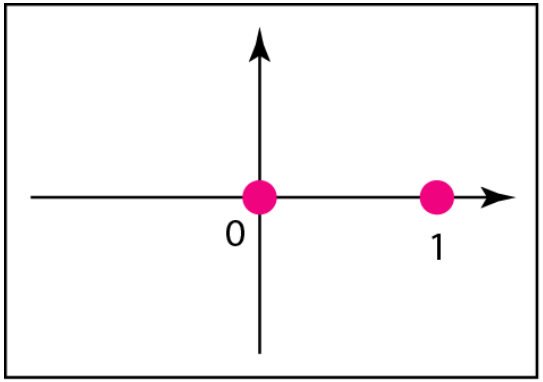
Solution



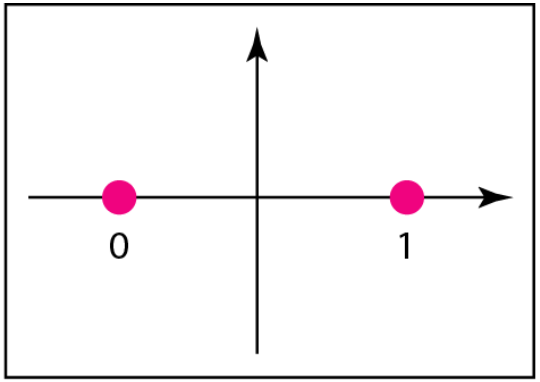
a. ASK (OOK) b. BPSK c. QPSK

ASK (OOK)

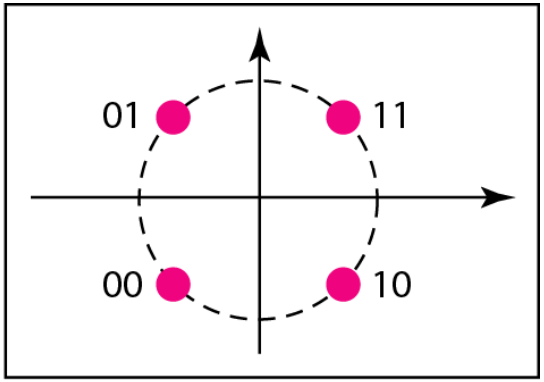
- *ASK uses only an in-phase carrier.*
- *Two points are on the X axis.*
- *Binary 0 has an amplitude of 0 V;*
- *Binary 1 has an amplitude of 1V.*



a. ASK (OOK)



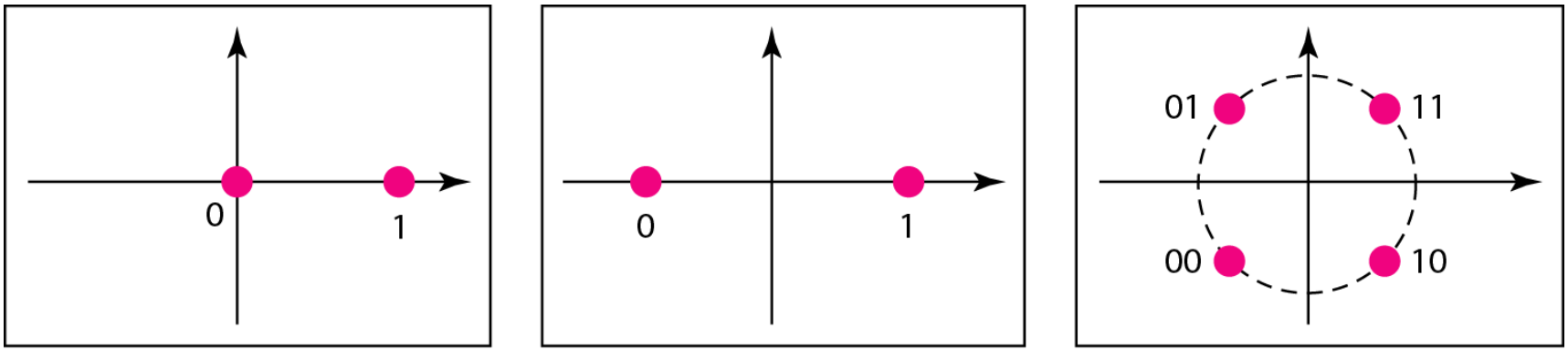
b. BPSK



c. QPSK

BPSK

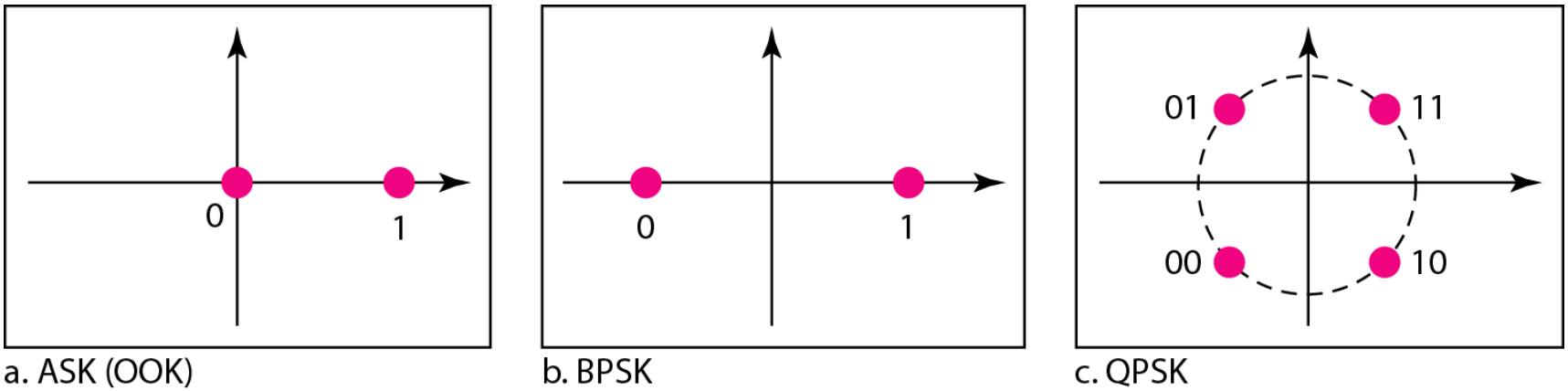
- BPSK uses only in-phase carrier.
- Polar NRZ signal for modulation.
  - Two signal types, one with amplitude "+1"and the other with amplitude "-1".
- BPSK has two different signals:
  - "1": Amplitude 1V and in phase.
  - "0": Amplitude 1V and 180^0 out of phase.



a. ASK (OOK) b. BPSK c. QPSK

QPSK

- QPSK uses two carriers: in-phase and quadrature.
- Point 11: two combined signals, both with an amplitude of 1 V. ("1" in in-phase carrier and "1" in quadrature carrier.)
- Amplitude of the composite 2-bit-data signal is $2^{1/2}$, and the phase is *45°.*
- *(45°, 135°, -135°, and -45°).*
- The amplitude of the carrier to be $1/(2^{1/2})$ can make the final amplitudes 1 V.



a. ASK (OOK) b. BPSK c. QPSK

Example 5.7

Find the bandwidth for a signal transmitting at 12 Mbps for QPSK. The value of d = 0.

Solution

For QPSK, 2 bits is carried by one signal element. This means that r = 2. So the signal rate (baud rate) is S = N × (1/r) = 6 Mbaud. With a value of d = 0, we have B = S = 6 MHz.

QAM (Quadrature Amplitude Modultion)

Quadrature amplitude modulation is a combination of ASK and PSK.

The minimum bandwidth required for QAM is the same as that for ASK and PSK.

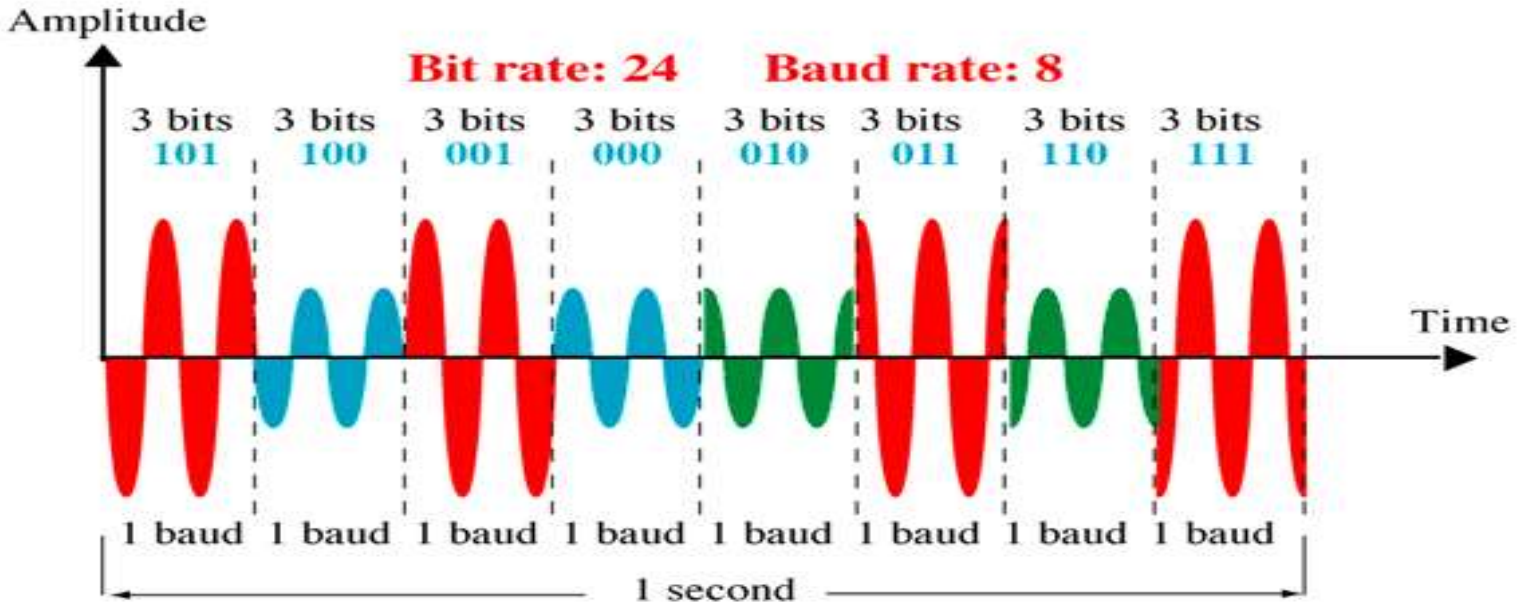


Figure 5.14 *Constellation diagrams for some QAMs*

- (a) Simplest 4-QAM using a unipolar NRZ (0,+V) to modulate each carrier. Same method for ASK (OOK).
- (b) 4-QAM using polar NRZ, exactly the same as QPSK.
- (c) 4-QAM using a signal with two positive levels for each carriers.
- (c) 16-QAM constellation of a signal with two positive and two negative levels for each carriers.

